

```
Windows PowerShell x Windows PowerShell x Windows PowerShell x + v
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W1103_MLflow_Model_Serving> notepad mlflow_model_serving.py
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W1103_MLflow_Model_Serving> python mlflow_model_serving.py
=====
W1103 - MLflow Model Serving & REST API
=====

Dataset:
Training samples : 120
Testing samples : 30
Features : 4

-----
EXPERIMENT 1
-----
Run ID : 17c5476734e54702b471fd436988d7cc
Parameters: {'n_estimators': 50, 'max_depth': 3, 'min_samples_split': 2}
Accuracy : 0.9667
Precision : 0.9697
Recall : 0.9667
F1 Score : 0.9666

-----
EXPERIMENT 2
-----
Run ID : 426ad7a408db4bae8d7da97e10e848ba
Parameters: {'n_estimators': 100, 'max_depth': 5, 'min_samples_split': 2}
Accuracy : 0.9333
Precision : 0.9333
Recall : 0.9333
F1 Score : 0.9333

-----
EXPERIMENT 3
-----
Run ID : a7c6d3acd27a4b088242d190f908e3d5
Parameters: {'n_estimators': 150, 'max_depth': 7, 'min_samples_split': 2}
Accuracy : 0.9000
```

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Windows PowerShell x Windows PowerShell x Windows PowerShell x + v

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EXPERIMENT 3
-----
Run ID : a7c6d3acd27a4b088242d190f908e3d5
Parameters: {'n_estimators': 150, 'max_depth': 7, 'min_samples_split': 2}
Accuracy : 0.9000
Precision : 0.9024
Recall : 0.9000
F1 Score : 0.8997

-----
EXPERIMENT 4
-----
Run ID : 5f44c2ff1d474eb2bc39efbc8d2c6ec8
Parameters: {'n_estimators': 200, 'max_depth': 10, 'min_samples_split': 2}
Accuracy : 0.9000
Precision : 0.9024
Recall : 0.9000
F1 Score : 0.8997

-----
EXPERIMENT 5
-----
Run ID : b0b2255b21204f0987afdeb8468d8ca0
Parameters: {'n_estimators': 250, 'max_depth': None, 'min_samples_split': 2}
Accuracy : 0.9000
Precision : 0.9024
Recall : 0.9000
F1 Score : 0.8997

=====
BEST MODEL
=====
Best Run ID : 17c5476734e54702b471fd436988d7cc
Best Accuracy: 0.9667
Best Parameters: {'n_estimators': 50, 'max_depth': 3, 'min_samples_split': 2}
```

```
Windows PowerShell x Windows PowerShell x Windows PowerShell x + v
Registering best model...
Successfully registered model 'W11D3_Iris_RandomForest'.
2026/09/17 16:38:26 WARNING mlflow.tracking._model_registry.fluent: Run with id 17c5476734e54702b471fd436988d7cc has no artifacts at
artifact path 'model', registering model based on models:/m-c55f11225eb64033a5403c61fe96472d instead
Created version '1' of model 'W11D3_Iris_RandomForest'.
Registered Model : W11D3_Iris_RandomForest
Model Version : 1

Loading registered model...
Registered model loaded successfully.
Loaded Model Accuracy: 0.9667

Sample Predictions:
Sample 1: setosa
Sample 2: virginica
Sample 3: versicolor
Sample 4: versicolor
Sample 5: setosa

=====
MODEL SERVING COMMAND
=====
mlflow models serve -m "models:/W11D3_Iris_RandomForest/1" -p 5001

REST endpoint:
POST http://127.0.0.1:5001/invocations

=====
W11D3 COMPLETED
=====

Completed:
[1] MLflow experiment created
[2] Parameters logged
[3] Metrics logged
[4] Model artifacts logged
```

```
Windows PowerShell x Windows PowerShell x Windows PowerShell x + v
Model Version : 1

Loading registered model...
Registered model loaded successfully.
Loaded Model Accuracy: 0.9667

Sample Predictions:
Sample 1: setosa
Sample 2: virginica
Sample 3: versicolor
Sample 4: versicolor
Sample 5: setosa

=====
MODEL SERVING COMMAND
=====
mlflow models serve -m "models:/W11D3_Iris_RandomForest/1" -p 5001

REST endpoint:
POST http://127.0.0.1:5001/invocations

=====
W11D3 COMPLETED
=====

Completed:
[1] MLflow experiment created
[2] Parameters logged
[3] Metrics logged
[4] Model artifacts logged
[5] Five experiments completed
[6] Best model identified
[7] Best model registered
[8] Registered model loaded
[9] Loaded model tested
[10] Model serving command generated
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving>
```

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station instructions
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving> mlflow models serve -m "models:/W11D3_Iris_RandomForest/1" -p 5001 --env-manager local
2026/09/17 16:44:10 INFO mlflow.models.flavor_backend_registry: Selected backend for flavor 'python_function'
2026/09/17 16:44:10 INFO mlflow.pyfunc.backend: === Running command 'uvicorn --host 127.0.0.1 --port 5001 --workers 1 mlflow.pyfunc.s
coring_server.app:app'
INFO: Started server process [12620]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://127.0.0.1:5001 (Press CTRL+C to quit)
2026/09/17 16:45:50 WARNING mlflow.pyfunc.scoring_server: If using 'instances' as input key, we internally convert the data type from
'records' (List[Dict]) type to 'list' (Dict[str, List]) type if the data is a pandas dataframe representation. This might cause sche
ma changes. Please use 'inputs' to avoid this conversion.

INFO: 127.0.0.1:63831 - "POST /invocations HTTP/1.1" 400 Bad Request
2026/09/17 16:46:18 WARNING mlflow.pyfunc.scoring_server: If using 'instances' as input key, we internally convert the data type from
'records' (List[Dict]) type to 'list' (Dict[str, List]) type if the data is a pandas dataframe representation. This might cause sche
ma changes. Please use 'inputs' to avoid this conversion.

INFO: 127.0.0.1:63832 - "POST /invocations HTTP/1.1" 400 Bad Request
INFO: 127.0.0.1:49678 - "POST /invocations HTTP/1.1" 200 OK

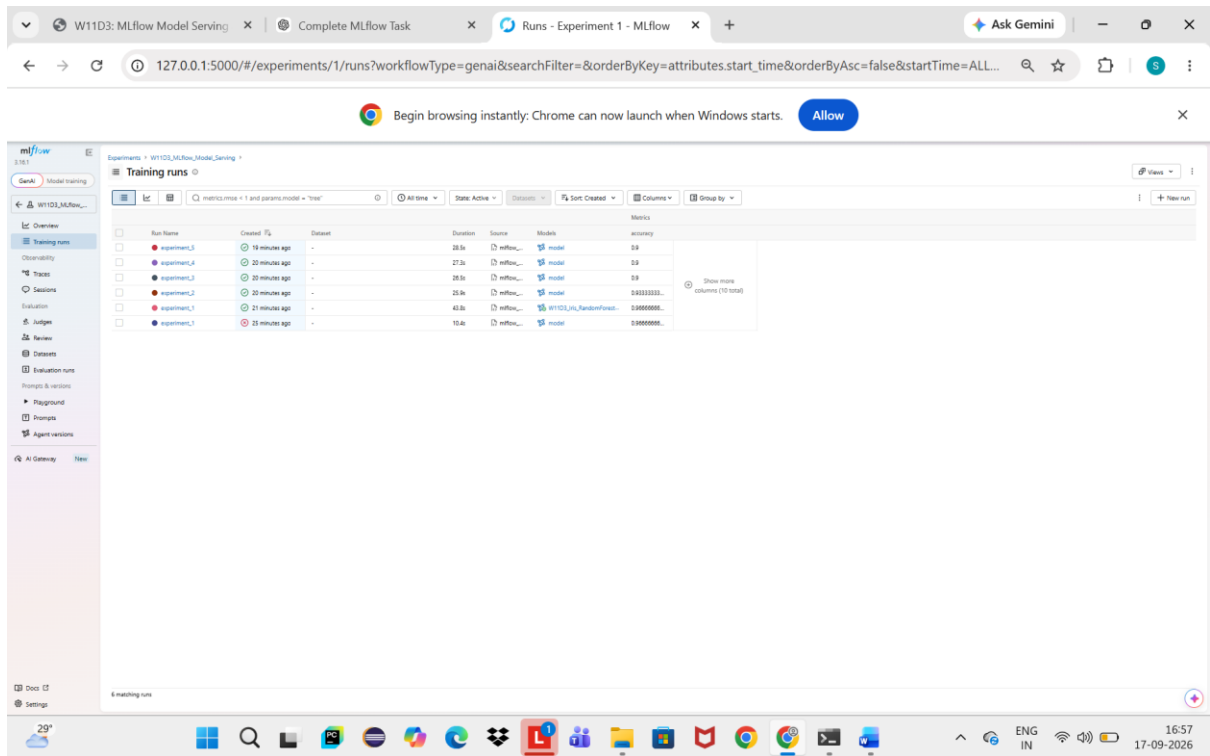
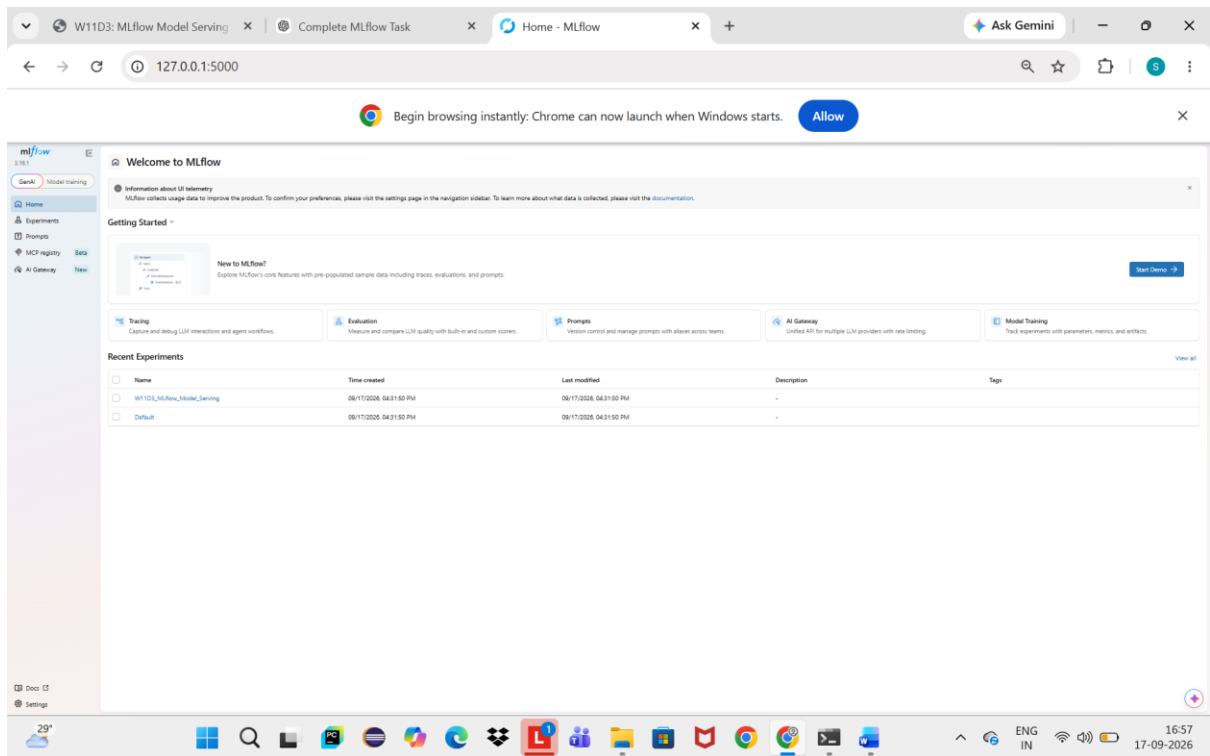
```

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Sanjana CA> cd "C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving"
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving> .\venv\Scripts\Activate.ps1
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving> mlflow ui --backend-store-uri "sqlite:
:///mlflow.db" --port 5000
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.
0 and configure --allowed-hosts and --cors-allowed-origins.
2026/09/17 16:54:39 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows sys
tem.). Server will start without job execution support. Errors will be surfaced at job invocation time.
2026/09/17 16:54:40 INFO: Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/09/17 16:54:40 INFO: Started parent process [16652]
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving\venv\Lib\site-packages\mlflow\server\fastapi_app.
py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to
https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_enviro
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving\venv\Lib\site-packages\mlflow\server\fastapi_app.
py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to
https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_enviro
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving\venv\Lib\site-packages\mlflow\server\fastapi_app.
py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to
https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_enviro
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving\venv\Lib\site-packages\mlflow\server\fastapi_app.
py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to
https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_enviro
2026/09/17 16:54:54 INFO: Started server process [27188]
2026/09/17 16:54:54 INFO: Waiting for application startup.
2026/09/17 16:54:54 INFO: Started server process [29148]
2026/09/17 16:54:54 INFO: Waiting for application startup.
2026/09/17 16:54:54 INFO: Started server process [18932]
2026/09/17 16:54:54 INFO: Waiting for application startup.
2026/09/17 16:54:54 INFO: Started server process [15772]

```



W11D3: MLflow Model Serving x Complete MLflow Task x Run 89bf1a18416c4b6a94c3dd1dbc2474a3

127.0.0.1:5000/#/experiments/1/runs/89bf1a18416c4b6a94c3dd1dbc2474a3

Begin browsing instantly: Chrome can now launch when Windows starts. [Allow](#)

mlflow 3.15.1

W11D3_MLflow_Model_Serving > Runs > experiment_1

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (6)

Search metrics

Metric	Value	Models
accuracy	0.9555555555555557	model
precision	0.9555555555555556	model
recall	0.9555555555555557	model
f1_score	0.9555555555555557	model

Parameters (6)

Search parameters

Parameter	Value
n_estimators	50
max_depth	3
min_sample_split	2
experiment_number	1

Logged models (1)

Model attributes						No dataset				
Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	precision	recall	f1_score
Output	0	model	Failed	25 minutes ago	-	-	0.9555555555555557	0.9555555555555556	0.9555555555555557	0.9555555555555557

Doc Settings

29° ENG IN 16:58 17-09-2026

W11D3: MLflow Model Serving x Complete MLflow Task x Run 17c5476734e54702b471fd436988d7cc

127.0.0.1:5000/#/experiments/1/runs/17c5476734e54702b471fd436988d7cc

Begin browsing instantly: Chrome can now launch when Windows starts. [Allow](#)

mlflow 3.15.1

W11D3_MLflow_Model_Serving > Runs > experiment_1

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (6)

Search metrics

Metric	Value	Models
accuracy	0.9555555555555557	model
precision	0.9555555555555556	model
recall	0.9555555555555557	model
f1_score	0.9555555555555557	model

Parameters (6)

Search parameters

Parameter	Value
experiment_number	1
n_estimators	50
max_depth	3
min_sample_split	2

Logged models (1)

Model attributes						No dataset				
Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	precision	recall	f1_score
Output	0	model	Ready	22 minutes ago	W11D3_MLflow_Model_Serving	-	0.9555555555555557	0.9555555555555556	0.9555555555555557	0.9555555555555557

Doc Settings

29° ENG IN 16:58 17-09-2026

W11D3: MLflow Model Serving x Complete MLflow Task x Run 426ad7a408db4bae8d7da x + Ask Gemini

127.0.0.1:5000/#/experiments/1/runs/426ad7a408db4bae8d7da97e10e848ba

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

mlflow 3.15.1 W11D3_MLflow_Model_Serving > Runs > experiment_2

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (6)

Metric	Value	Models
accuracy	0.9333333333333333	model
precision	0.9333333333333333	model
recall	0.9333333333333333	model
f1_score	0.9333333333333333	model

Parameters (6)

Parameter	Value
experiment_number	2
n_estimators	100
max_depth	5
min_sample_split	2

Logged models (1)

Type	Step	Model name	Status	Created	Registered models	Dataset	No dataset accuracy	precision	recall	f1_score
Output	0	model	Ready	22 minutes ago	-	-	0.9333333333333333	0.9333333333333333	0.9333333333333333	0.9333333333333333

About this run

Created at 08/11/2026 04:36:37 PM

Created by Sargana CA

Experiment ID 1

Status Finished

Run ID a7c6d3acd27a4b088242d190f908e3d5

Duration 25.5s

Source mlflow_model_serving.py

Registered prompts -

Datasets None

Tags Add tags

Registered models None

29° 16:59 17-09-2026

W11D3: MLflow Model Serving x Complete MLflow Task x Run a7c6d3acd27a4b088242d190f908e3d5 x + Ask Gemini

127.0.0.1:5000/#/experiments/1/runs/a7c6d3acd27a4b088242d190f908e3d5

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

mlflow 3.15.1 W11D3_MLflow_Model_Serving > Runs > experiment_3

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (6)

Metric	Value	Models
accuracy	0.9	model
precision	0.90236602366024	model
recall	0.9	model
f1_score	0.8997460774635839	model

Parameters (6)

Parameter	Value
experiment_number	3
n_estimators	150
max_depth	7
min_sample_split	2

Logged models (1)

Type	Step	Model name	Status	Created	Registered models	Dataset	No dataset accuracy	precision	recall	f1_score
Output	0	model	Ready	22 minutes ago	-	-	0.9	0.90236602366024	0.9	0.8997460774635839

About this run

Created at 08/11/2026 04:37:03 PM

Created by Sargana CA

Experiment ID 1

Status Finished

Run ID a7c6d3acd27a4b088242d190f908e3d5

Duration 25.5s

Source mlflow_model_serving.py

Registered prompts -

Datasets None

Tags Add tags

Registered models None

29° 16:59 17-09-2026

W11D3: MLflow Model Serving x Complete MLflow Task x Run 5f44c2ff1d474eb2bc39efbc8 x + Ask Gemini

127.0.0.1:5000/#/experiments/1/runs/5f44c2ff1d474eb2bc39efbc8d2c6ec8

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

mlflow 3.15.1 W11D3_MLflow_Model_Serving > Runs > experiment_4

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (6)

Metric	Value	Models
accuracy	0.9	model
precision	0.9023568023568024	model
recall	0.9	model
f1_score	0.8997468734335839	model

Parameters (6)

Parameter	Value
experiment_number	4
n_estimators	200
max_depth	10
min_sample_split	2

Logged models (1)

Model attributes					No dataset					
Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	precision	recall	f1_score
Output	0	model	Ready	22 minutes ago	-	-	0.9	0.9023568023568024	0.9	0.8997468734335839

Doc Settings

29° ENG IN 16:59 17-09-2026

W11D3: MLflow Model Serving x Complete MLflow Task x Run b0b2255b21204f0987afdeb8468d8ca0 x + Ask Gemini

127.0.0.1:5000/#/experiments/1/runs/b0b2255b21204f0987afdeb8468d8ca0

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

mlflow 3.15.1 W11D3_MLflow_Model_Serving > Runs > experiment_5

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (6)

Metric	Value	Models
accuracy	0.9	model
precision	0.9023568023568024	model
recall	0.9	model
f1_score	0.8997468734335839	model

Parameters (6)

Parameter	Value
experiment_number	5
n_estimators	250
max_depth	None
min_sample_split	2

Logged models (1)

Model attributes					No dataset					
Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	precision	recall	f1_score
Output	0	model	Ready	22 minutes ago	-	-	0.9	0.9023568023568024	0.9	0.8997468734335839

Doc Settings

29° ENG IN 17:00 17-09-2026

```

P:\Cynaris\W11D3_MLflow_Model_Serving> Invoke-WebRequest -Uri "http://127.0.0.1:5001/invocations" -Method Post -ContentType "application/json" -Body '{"dataframe_records":[{"sepal length (cm)":5.1,"sepal width (cm)":3.5,"petal length (cm)":1.4,"petal width (cm)":0.2},{"sepal length (cm)":6.7,"sepal width (cm)":3.1,"petal length (cm)":4.7,"petal width (cm)":1.5},{"sepal length (cm)":6.3,"sepal width (cm)":3.3,"petal length (cm)":6.0,"petal width (cm)":2.5}]}'
(venv) PS C:\Users\Sanjana_CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving> Invoke-RestMethod `
>> -Uri "http://127.0.0.1:5001/invocations"
>> -Method Post
>> -ContentType "application/json"
>> -Body '{"dataframe_records":[{"sepal length (cm)":5.1,"sepal width (cm)":3.5,"petal length (cm)":1.4,"petal width (cm)":0.2},{"s
epal length (cm)":6.7,"sepal width (cm)":3.1,"petal length (cm)":4.7,"petal width (cm)":1.5},{"sepal length (cm)":6.3,"sepal width (c
m)":3.3,"petal length (cm)":6.0,"petal width (cm)":2.5}]}'

predictions
-----
{0, 1, 2}

(venv) PS C:\Users\Sanjana_CA\OneDrive\Documents\CynarisInternship\W11D3_MLflow_Model_Serving>
```